

B. M. S. College of Engineering, Bengaluru - 560019

Autonomous Institute Affiliated to VTU

JAN / FEB – 2021 Grade Improvement Examinations of Even Semester

Programme: B.E.

Branch : Computer Science and Engineering

Course Code: 19CS4PCADA

Course: Analysis and Design of Algorithms

Semester: IV

Duration: 3 hrs.

Max Marks: 100

Date: 19.02.2021

Instructions to Candidates (Shall Include instruction regarding issue of Charts / Tables etc.,)

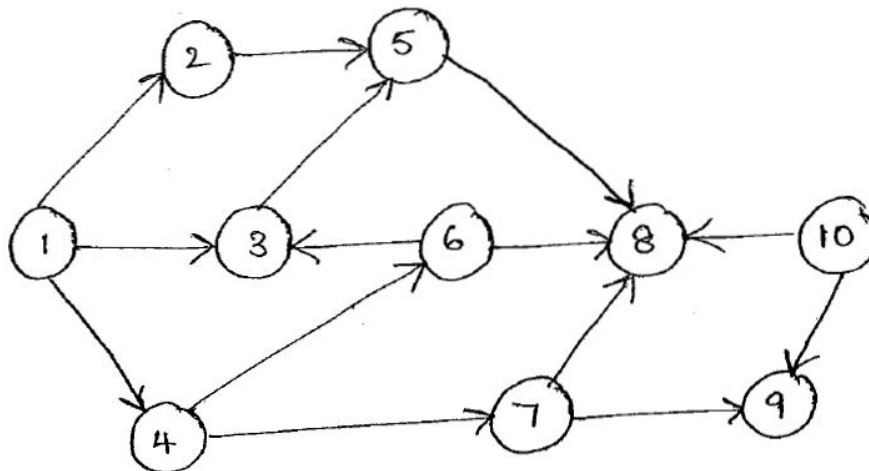
Q No. Marks

Unit-1

- 1a) Analyze the code given below and find the time complexity: 06
- ```
void function(int n)
{
 int count = 0;
 for (int i=n/2; i<=n; i++)
 for (int j=1; j<=n; j = 2 * j)
 for (int k=1; k<=n; k = k * 2)
 count++;
}
```
- 1b) Write an algorithm to find the GCD of an array of numbers. 06
- 1c) Consider the following recurrence relation: 08
- $T(n) = 2 * T(\sqrt{n}) + 1, T(1) = 1$
- Solve the above recurrence relation to obtain the time complexity.

## Unit-2

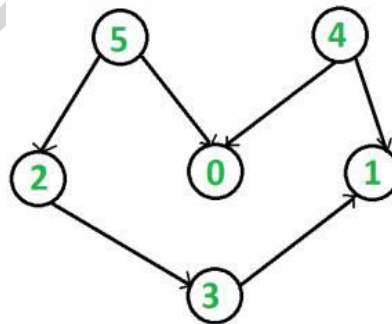
- 2a) Obtain DFS and BFS traversal for the following graph. Neatly demonstrate all the steps. 08



- 2b) The recurrence  $T(n) = 7T(n/2) + n^2$  describes the running time of an algorithm 06  
 A. Solve the recurrence relation for  $T(n)$  using master's theorem. A competing algorithm A' has running time of  $T'(n) = aT(n/4) + n^2$ . What is the largest value such that A' is asymptotically faster than A?
- 2c) Develop an algorithm for Brute-force string matching and also derive the time 06  
 complexity for the same.

OR

- 3a) Write an algorithm for the following scenario: 10  
 Given a connected graph, check if the graph is bipartite or not.  
 A Bipartite Graph is a graph whose vertices can be divided into two independent sets, U and V such that every edge (u, v) either connects a vertex from U to V or a vertex from V to U. In other words, for every edge (u, v), either u belongs to U and v to V, or u belongs to V and v to U. We can also say that there is no edge that connects vertices of same set.  
 A bipartite graph is possible if the graph coloring is possible using two colors such that vertices in a set are colored with the same color.  
 Also, obtain the time complexity for the same.
- 3b) Write an algorithm for the following scenario: 10  
 Given a directed graph with N vertices and M edges that may contain cycles, the task is to find the lexicographically smallest topological ordering of the graph if it exists otherwise print -1 (if the graph has cycles).  
 Lexicographically smallest topological ordering means that if two vertices in a graph do not have any incoming edge then the vertex with the smaller number should appear first in the ordering.  
 For Example, in the graph below some of the possible topological orderings 5 2 3 4 0 1, 5 0 2 4 3 1, etc.  
 But the smallest ordering is 4 5 0 2 3 1.



### Unit-3

- 4a) Analyze and derive the time complexity for Quick sort algorithm for the following scenario: 05  
The array is already sorted and the first element in the array is chosen as the pivot element.
- 4b) Build a heap and also apply heap sort technique to sort the following numbers in ascending order: 10  
25, 57, 48, 37, 12, 92, 86, 33
- 4c) Demonstrate in terms of time complexity, how Karatsuba algorithm for multiplication of two large integers using Divide and Conquer technique is efficient? 05

### Unit-4

- 5a) Given a rod of length 'n' inches and an array of prices that contains prices of all pieces of size smaller than 'n'. Determine the maximum value obtainable by cutting up the rod and selling the pieces. For example, if length of the rod is 8 and the values of different pieces are given as following, then the maximum obtainable value is 22 (by cutting in two pieces of lengths 2 and 6) 08

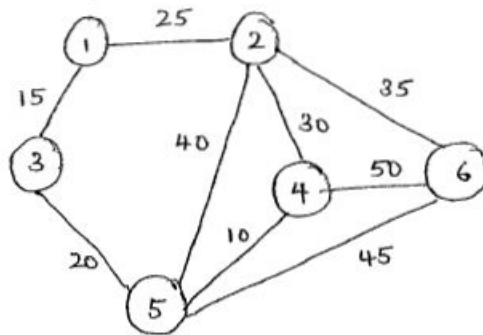
|        |  |   |   |   |   |    |    |    |    |
|--------|--|---|---|---|---|----|----|----|----|
| length |  | 1 | 2 | 3 | 4 | 5  | 6  | 7  | 8  |
| -----  |  |   |   |   |   |    |    |    |    |
| price  |  | 1 | 5 | 8 | 9 | 10 | 17 | 17 | 20 |

And if the prices are as following, then the maximum obtainable value is 24 (by cutting in eight pieces of length 1)

|        |  |   |   |   |   |    |    |    |    |
|--------|--|---|---|---|---|----|----|----|----|
| length |  | 1 | 2 | 3 | 4 | 5  | 6  | 7  | 8  |
| -----  |  |   |   |   |   |    |    |    |    |
| price  |  | 3 | 5 | 8 | 9 | 10 | 17 | 17 | 20 |

Design an algorithm for the above scenario using Dynamic Programming.

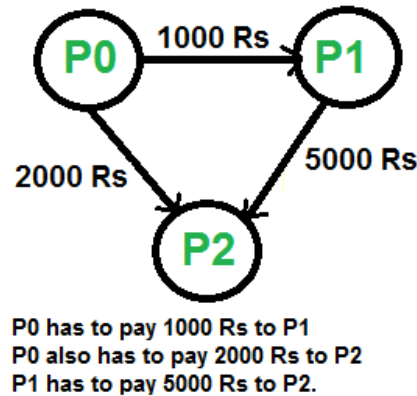
- 5b) Apply Prim's algorithm to find the Minimum Spanning Tree of the following graph: 08



- 5c) Given a number of friends who have to give or take some amount of money from one another. Design an algorithm by which the total cash flow among all the friends is minimized. 04

Example:

Following diagram shows input debts to be settled.



For the above example, show the optimal way of settling the debts in the form of a graph only.

OR

- 6a) Given  $n$  dice each with  $m$  faces, numbered from 1 to  $m$ , find the number of ways to get sum  $X$  using Greedy technique.  $X$  is the summation of values on each face when all the dice are thrown. 10  
For example, let there be 3 dice, each with 6 faces and we need to find the number of ways to get sum 8. The required answer is 21.

- 6b) Given weights and values of ' $n$ ' items, we need to put these items in a knapsack of capacity  $W$  to get the maximum total value in the knapsack. 10

In the 0-1 Knapsack problem, we are not allowed to break items. We either take the whole item or don't take it.

Input:

Items as (value, weight) pairs

$arr[] = \{\{60, 10\}, \{100, 20\}, \{120, 30\}\}$

Knapsack Capacity,  $W = 50$ ;

Output:

Maximum possible value = 220

by taking items of weight 20 and 30 kg

In Fractional Knapsack, we can break items for maximizing the total value of knapsack. This problem in which we can break an item is also called the fractional knapsack problem.

Input :

Same as above

Output :

Maximum possible value = 240

By taking full items of 10 kg, 20 kg and

2/3rd of last item of 30 kg

Design a Greedy Algorithm for the above scenario of fractional Knapsack problem. Also, derive the time complexity for the same.

### Unit-5

- 7a) A Maze is given as  $N \times N$  binary matrix of blocks where source block is the upper left most block i.e.,  $\text{maze}[0][0]$  and destination block is lower rightmost block i.e.,  $\text{maze}[N-1][N-1]$ . A rat starts from source and has to reach the destination. The rat can move only in two directions: forward and down. In the maze matrix, 0 means the block is a dead end and 1 means the block can be used in the path from source to destination. Note that this is a simple version of the typical Maze problem.

Following is an example maze.

Gray blocks are dead ends (value = 0).

|        |  |  |       |
|--------|--|--|-------|
| Source |  |  |       |
|        |  |  |       |
|        |  |  |       |
|        |  |  | Dest. |

Following is binary matrix representation of the above maze.

{1, 0, 0, 0}

{1, 1, 0, 1}

{0, 1, 0, 0}

{1, 1, 1, 1}

Following is a maze with highlighted solution path.

|        |  |  |       |
|--------|--|--|-------|
| Source |  |  |       |
|        |  |  |       |
|        |  |  |       |
|        |  |  | Dest. |

Following is the solution matrix (output) for the above input matrix.

{1, 0, 0, 0}  
{1, 1, 0, 0}  
{0, 1, 0, 0}  
{0, 1, 1, 1}

All entries in solution path are marked as 1.

Design an algorithm for the above scenario using Backtracking technique.

- 7b) Differentiate between NP-complete and NP-hard problems with an example for each. 05
- 7c) Demonstrate the state-space tree for the following scenario: 05  
Given the set  $S=\{1,2,5,6,8\}$  and the required sum value  $d=9$ . Find the possible subsets with sum value= $d$  using backtracking.